SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with 'A+' Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23	IV / IV - B.Tech. I - Semester								
ELECTRICAL & ELECTRONICS ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23EE4101	Electric Vehicles	**PC**	3	0	0	3	30	70	100
B23HS4105	Energy Management and Audit	**HS**	2	0	0	2	30	70	100
#PE-IV	Professional Elective - IV	**PE**	3	0	0	3	30	70	100
#PE-V	Professional Elective - V	**PE**	3	0	0	3	30	70	100
#OE-III	Open Elective – III	**OE**	3	0	0	3	30	70	100
#OE-IV	Open Elective – IV	**OE**	3	0	0	3	30	70	100
B23EE4110	Electric Vehicles Laboratory	**SEC**	0	1	2	2	30	70	100
B23MC4102	Technical Paper Writing & IPR	**MC**	2	0	0	-	30	-	30
B23EE4111	Evaluation of Industry Internship /Mini Project.	**PR**	-	-	-	2	--	50	50
		TOTAL	**19**	**1**	**2**	**21**	**240**	**540**	**780**

	Course Code	Course
#PE-IV	B23EE4102	Battery Management &Charging Systems
	B23EE4103	Power System Operation & Control
	B23EE4104	Autonomous Vehicle Systems
	B23EE4105	MOOCS - IV
#PE-V	B23EE4106	Power Electronic Drives
	B23EE4107	Power System Protection
	B23EE4108	Advanced Control Systems
	B23EE4109	MOOCS - V
#OE-III	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE ME or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE or ME or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4101	PC	3	--	--	3	30	70	3 Hrs.

ELECTRIC VEHICLES

(For EEE)

Course Objectives: Students will learn about

1.	The introductory concepts of EVs and dynamic modelling equations of EVs.
2.	The various configurations of EVs and HEVs and power train components.
3.	The various Energy storage systems for EVs and understand their characteristics.
4.	The Drive systems of EVs and their control.
5.	The charging technologies in EVs.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Analyze** and understand dynamic modelling and design considerations of Electrical vehicles.	**K4**
2.	**Illustrate** the architecture of electric vehicles and power train components	**K3**
3.	**Evaluate** battery performance parameters for EVs and understand other energy storage methods for EVs	**K4**
4.	**Analyze** and understand the electric drives using power electronic converters for EVs.	**K4**
5.	**Illustrate** the EV charger infrastructure.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Introduction To Electric Vehicles And Modeling** Introduction to Electric Vehicles (EV), Hybrid Electric Vehicles (HEV), EV History, EV Advantages, overall Comparison of EV with Internal Combustion Engine vehicles, feasibility of EV, Vehicle Mechanics and Dynamics modelling-Roadway Fundamentals, Laws of Motion, Vehicle Load Forces, Vehicle Kinetics, Dynamics of Vehicle Motion, Propulsion Power, Force-Velocity Characteristics, Maximum Gradeability, Velocity and Acceleration for Constant Tractive Force on Level Road, General Acceleration for Non-constant Tractive Force, Propulsion System Design, Design Considerations.
UNIT-II (10 Hrs)	**Architecture Of EVs And Power Train Components** Architecture of EVs – 2-motors, 4 motors, and hub motor, Architecture of Hybrid EVs – Series, Parallel, Series parallel, complex, Plug-in Hybrid Electric Vehicles (PHEV), Architecture of Fuel cell EV, Power train components of EVs - EV Transmission Configurations, Transmission Components, Ideal Gearbox: Steady State Model, and EV Motor Sizing, Standard Drive Cycles
UNIT-III (10 Hrs)	**Energy Storage For EV** Battery- Battery Basics, Different types, Lead Acid Batteries and Lithium Batteries (Li-ion,

	Li-Polymer), Battery Parameters, Battery Power, Battery modelling, Battery Failure and Protection, Battery Management system, Battery Pack Design, Lifetime and Sizing, Fuel cell, Hydrogen Storage Systems, Ultra capacitors, Flywheel, Battery Recycling.
UNIT-IV (10 Hrs)	**Electric Vehicle Motor Drives** Electric Drive Components of EV, Requirement of electric motors for EV, Permanent Magnet Synchronous Motor (PMSM) Drive - PMSM motor operation using simple controller, Brushless DC (BLDC) Motor Drive -BLDC motor operation, model, DC link current control, Comparison of PMSM and BLDC, Switched Reluctance Motor (SRM) Drive - SRM motor operation, Converter topologies for SRM.
UNIT-V (10 Hrs)	**EV Charging Technology** Overview of the EV battery charging system, Infrastructure Needed for Charging Electric Vehicles, Basic Requirements for Charging System, Charger Architectures-AC charger, DC Charger, Basics of Wireless charging – Static and Dynamic charging, EV Charging Standards and Technologies, Effects of EV load on the Grid, Introduction to V2G and V2V technologies.

Textbooks:

1.	Iqbal Husain, “Electric and Hybrid Vehicles: Design Fundamentals”, CRC Press, Taylor & Francis Group, 2003.
2.	John G. Hayes and G.A. Goodarzi, “Electric Power train - Energy Systems, Power electronics and drives for Hybrid, electric and fuel cell vehicles” Wiley Publication, 1st edition, 2018.

Reference Books:

1.	James Larminie, John Lowry, “Electric Vehicle Technology Explained” Wiley publication, 2^{nd} Edition, 2012.
2.	M. Ehsani, Y. Gao, and A. Emadi, “Modern Electric, Hybrid Electric, and Fuel Cell Vehicles”, CRC Press, 2010.

e-Resources

1.	https://nptel.ac.in/courses/108106170
2.	https://nptel.ac.in/courses/108106182

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4105	HS	3	--	--	3	30	70	3 Hrs.

ENERGY MANAGEMENT AND AUDIT

(For EEE)

Course Objectives: Students will learn about

1.	The Energy scenario across the globe.
2.	The Principles of Energy Management, Qualities and functions of energy manager.
3.	The Power factor control, Tariff, Energy Efficient Lighting, Life Cycle Cost Analysis
4.	The DSM and Load management.
5.	The Objectives and types of Energy Audit, Energy Conservation Schemes

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Explore the energy scenario across the globe.	K3
2.	Explore the Principles of Energy management and various duties of energy manager	K3
3.	Explore energy efficient systems and Life Cycle Cost Analysis	K3
4.	Describe DSM and Load management	K3
5.	Explore the methods and concept of energy audit	K3

SYLLABUS

UNIT-I (10Hrs)	**ENERGY SCENARIO**: Introduction, World Energy Sources and their Classification, Coal as an Energy Source, Oil as an Energy Source, Natural Gas as an Energy Source, Global Scenario for Non-Renewable Energy Sources, Global Scenario for Renewable Energy Sources, Indian Scenario for Non-Renewable Energy Sources, Indian Scenario for Renewable Energy Sources, Need for Energy Sector Reforms in India, Energy Planning, Energy Utilization Patterns, Strategy for Meeting Future Energy Requirements.
UNIT-II (10 Hrs)	**Energy Management:** Introduction, Scope of energy management, Necessary steps of Energy Management Programme, Principles of Energy Management, Qualities and functions of Energy manager, Indian Energy Conservation Act 2022 (EC Act), The Electricity Act 2003.
UNIT-III (10 Hrs)	**Electrical Energy Management:** Introduction, Power factor control, Tariff, Energy efficient motors, Case Study, Energy Efficient Lighting, Life Cycle Energy Analysis, Life Cycle Cost Analysis, Comparison between Simple Pay Back & Life Cycle Cost Analysis, Equivalent Annual Worth (EAW), Break even Analysis.
UNIT-IV (10 Hrs)	**Demand Side Management and Energy Instruments:** Introduction to DSM, concept of DSM, benefits of DSM, Different techniques of DSM,

	time of day pricing, multi-utility power exchange model, time of day models for planning. Load management - load priority technique, peak clipping, peak shifting, valley filling, strategic conservation, Energy Instruments, data loggers, thermocouples, pyrometers, lux meters, tongue testers.
UNIT-V (10 Hrs)	**Energy Auditing**: Introduction, Objectives of Energy Audit, Control of Energy, Uses of Energy, Energy Conservation Schemes, Energy Index, Cost Index, Pie Chart, Sankey Diagram, Load Profile, Types of Energy Audit, General Energy Audit, Energy Audit Case Studies
Textbooks:	
1.	Energy Conservation and Management by S. S. Thipse, Alpha Science International, Limited, 1st Edition, 2014.
2.	Energy Management and Conservation by K.V. Sharma & P.Venkatasesaiah, I.K. International Pvt. Ltd., New Delhi, 2nd Edition, 2020.
Reference Books:	
1.	Energy management by W.R. Murphy & G. Mckay Butter worth, Elsevier publications, 2nd Edition, 2009.
2.	Energy management hand book by W.C.Turner, John wiley and sons, 9th Edition, 2020.
e-Resources	
1.	https://nptel.ac.in/courses/112105221
2.	https://nptel.ac.in/courses/109106161

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4102	PE	3	--	--	3	30	70	3 Hrs.

BATTERY MANAGEMENT & CHARGING SYSTEMS

(For EEE)

Course Objectives: Students will learn about	
1.	To understand the need, functions and architecture of Battery Management Systems (BMS)
2.	To study the battery parameters and modelling approaches used in BMS
3.	To comprehend the state estimation measurement, protection, balance, communication and topologies of BMS
4.	To impart knowledge on charging methods including CC, CV, CP, pulse, and off-board charging circuit designs.
5.	To understand the charging infrastructure and its various types

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Apply the concepts of Battery Management System to explain its need, working principles, architecture and real-world implementation using Microprocessor based BMS	K3
2.	Analyze battery parameters, modeling methods, and evaluate SoC (Coulomb counting), DoD, SoH, and C-rate.	K3
3.	Analyze BMS functions (protection, balancing, thermal management, communication) and compare different BMS topologies (centralized, modular, master–slave, distributed)..	K3
4.	Analyze and compare various EV charging technologies, charging methods, and evaluate the operation of on-board (Level 1 & 2) and off-board (Level 3) charging circuits.	K3
5.	Analyze different types of EV charging infrastructure including domestic, public, fast charging stations and battery swapping systems	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Battery Management System (BMS):** Battery Management System (BMS), why a Battery Needs a BMS, how a BMS Makes a Storage System Efficient, Safe, and Dependable, example of a BMS in a Real System - Microprocessor based BMS, System failures due to the absence of a BMS,basic functions of a BMS, General development flow of the power battery management system.
UNIT-II (10 Hrs)	**Battery Parameters and Modelling:** Battery parameters - Capacities, Measurement of Voltage, Temperature & Current, Battery modelling: Equivalent-circuit models (ECMs), Physics-based models (PBMs), Empirical modeling approach, Physics-based modelling approach. State of Charge (SoC),SoC estimation of a Battery-Coulomb Counting Method. Depth of Discharge (DoD), State of

	Health (SoH), C rate.
UNIT-III (10 Hrs)	**Topologies & communication in BMS:** Protection, Balancing/Balancing Control & Thermal Management, Evaluation, Communication Interface for BMS. Topological Structure of a BMS - Centralized, Modular, Master-Slave, Distributed & their comparison
UNIT-IV (10 Hrs)	**Charging Technologies:** Introduction, Types and Technology of EV Charging, Comparison between different power level types for EV charging, Topologies for EV Charging, Charging methods -Semi-constant current, constant current (CC), Constant voltage (CV), constant power (CP), pulse charging, and trickle charging. Charging Circuits-On-Board Charger (Level 1 and Level 2 Chargers),Off-Board Charger (Level 3)
UNIT-V (10 Hrs)	**Charging Infrastructure:** Domestic Charging Infrastructure, Public charging Infrastructure, Normal Charging Station,Occasional Charging Station, Fast Charging Station, Battery Swapping Station, Move-and chargezone

Textbooks:	
1.	Xiaojun Tan, Andrea Vezzini, Yuqian Fan, Neeta Khare, You Xu, Liangliang Wei, “Battery Management System and Its Applications”, ISBN: 9781119154020, 1119154022,Publisher: John Wiley & Sons, Incorporated, © 2023 China Machine Press
2.	Ashutosh K. Giri Madhusudan Singh “Electric Vehicle Charging Infrastructures and its Challenges” ISSN 2730-6453, Springer Nature Singapore PteLtd.2025,https://doi.org/10.1007/978-981-96-0361-9
Reference Books:	
1.	Kundan Kumar, Ambrish Devanshu, and Sanjeet K. Dwivedi, “Electric Vehicle Propulsion Drives and Charging Systems” 1st Edition, ISBN-13978-10325281, Publisher: CRC Press, 2024
2.	Davide Andrea, “Battery Management Systems for Large Lithium-Ion Battery Packs”, Power engineering series, Artech House Publishers, 2010,
3.	Nicolae Tudoroiu “Battery Management Systems of Electric and Hybrid Electric Vehicles” Engineering - Energy & Power Resources, MDPI AG, 2021, ISBN 13: 9783036510606
e-Resources	
1.	https://nptel.ac.in/courses/108106170
2.	https://nptel.ac.in/courses/108107715

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4103	PE	3	--	--	3	30	70	3 Hrs.

POWER SYSTEM OPERATION AND CONTROL

(For EEE)

Course Objectives: Students will learn about

1.	The optimal dispatch of generation with and without losses
2.	The optimal scheduling of hydro thermal systems and unit commitment
3.	The load frequency control for single area system with and without controllers.
4.	The load frequency control for two area system with Tie-line bias, Economic dispatch control, automatic voltage control, generator constraints and governor dead band.
5.	The stability enhancement methods, preventive & emergency control.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Compute the economic load scheduling for Thermal power plants.	**K3**
2.	Illustrate the concepts of hydro thermal systems and unit commitment.	**K3**
3.	Analyze the frequency deviations of a single area power system	**K4**
4.	Analyze the Load frequency control of a two-area system with tie-line bias and illustrate the concepts of automatic voltage control, generator constraints and governor dead band.	**K4**
5.	Explore stability enhancement methods, preventive & emergency control of power system.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**ECONOMIC OPERATION OF POWER SYSTEMS:** Optimal operation of Generators in Thermal power stations, Heat rate curve, Cost Curve, Incremental fuel and Production costs, Input–output characteristics, Optimum generation allocation with line losses neglected, Optimum generation allocation including the effect of transmission line losses, Loss Coefficients, General transmission line loss formula.
UNIT-II (10Hrs)	**HYDROTHERMAL SCHEDULING:** Optimal scheduling of Hydrothermal System: Mathematical formulation – Solution technique for short term hydrothermal scheduling problem. **OPTIMAL UNIT COMMITMENT:** Need for unit commitment, Constraints in unit commitment, Cost function formulation, Solution methods, Priority ordering, Dynamic programming.
UNIT-III (10Hrs)	**LOAD FREQUENCY CONTROL-I:** Modelling of steam turbine, Generator, Modelling of speed governing system, Transfer function, Necessity of keeping frequency constant. Definitions of Control area, Single area control system, Block diagram representation of an isolated power system, Steady state

	analysis, Dynamic response, Uncontrolled case. Proportional plus Integral control of single area and its block diagram representation, Steady state response.
UNIT-IV (10Hrs)	**LOAD FREQUENCY CONTROL-II:** Two area load frequency control and its block diagram, Load Frequency Control of two area system uncontrolled case and tie line bias controlled case. Load frequency control and economic dispatch control. Automatic voltage regulator (AVR).
UNIT-V (10Hrs)	**PREVENTIVE AND EMERGENCY CONTROL:** Power system operating states, Equality and Inequality constraints, Concepts of preventive and emergency control, Coherent area dynamics, Stability enhancement methods, long term frequency dynamics, Average system frequency, Centre of inertia. Factors affecting the power system security.
Textbooks:	
1.	Power Generation - Operation and Control by Allen J Wood - Bruce F Wollen Berg 3rd Edition - Wiley Publication 2014.
2.	Electrical Energy Systems Theory - by O.I.Elgerd, Tata McGraw-Hill Publishing Company Ltd, 2nd edition.
3.	Modern Power System Analysis - by I.J.Nagrath & D.P.Kothari, Tata McGraw-Hill Publishing Company ltd, 2nd edition.
Reference Books:	
1.	Power System Analysis and Stability by S.S.Vadhera - Khanna Publications - 4th edition - 2005.
2.	Power System Analysis by HadiSadat, Third Edition, Tata McGraw Hill publication-2010.
3.	Power System stability & control - Prabha Kundur - TMH - 1994.
e-Resources	
1.	https://nptel.ac.in/courses/108104191
2.	https://nptel.ac.in/courses/108104052

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4104	PE	3	--	--	3	30	70	3 Hrs.

AUTONOMOUS VEHICLE SYSTEMS

For EEE

Course Objectives: Students will learn about

1.	The core principles of autonomous vehicles (AV), including their evolution, architecture, and fundamental kinematics/dynamics.
2.	The sensor technologies and algorithms that enable AVs to interpret their surroundings.
3.	The vehicle behavior and implement control strategies for safe navigation.
4.	Decision-making frameworks for complex driving scenarios.
5.	The Robot Operating System (ROS) concepts for AV applications.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Understand AV basics, autonomy levels, and apply AV kinematics.	K3
2.	Explore sensor technologies and how AVs perceive environments.	K4
3.	Model vehicle dynamics and control for stable driving.	K4
4.	Design decision-making systems for smart autonomous behavior.	K5
5.	Implement AV algorithms using ROS 2	K3

SYLLABUS

UNIT-I (10Hrs)	**Vehicle Autonomy Essentials** Introduction to AVs, historical evolution, SAE autonomy levels, AV architecture–Sensors, perception, planning, control, and decision-making, Reference frames, coordinate systems, vehicle kinematics, and dynamics, Transformation techniques–rotation matrices, homogeneous transforms, 3D representations.
UNIT-II (10 Hrs)	**Vehicle Dynamics and Motion Control** Longitudinal and Lateral dynamics, Dynamic bicycle model, Actuation Systems—Steering system dynamics, Brake-by-wire and throttle control, State-space representations, Motion Control, Control Objectives—Path tracking accuracy, Stability criteria (yaw rate, slip angle), Ride comfort metrics, Control Methods— PID control for basic maneuvers.
UNIT-III (10 Hrs)	**Sensor Systems and Perception** Sensors: LiDAR, Radar, Cameras, IMU, Ultrasonics, GNSS, Sensor fusion, State estimation–Kalman Filtering, Simultaneous Localization and Mapping (SLAM), object tracking, Visual odometry, LiDAR-based Mapping.
UNIT-IV (10 Hrs)	**Planning and Intelligent Decision-Making** Planning Hierarchy—Behavior planning vs motion planning, Scenario-based requirements,

	Rule-Based Systems—Finite State Machines (FSM) for driving behaviors, Decision trees for maneuver selection, Learning-Based Systems—Markov Decision Processes (MDP) framework, Deep Q-Networks (DQN) for driving policies, Reward Engineering—Safety metrics, Comfort parameters, Efficiency objectives.
UNIT-V (10 Hrs)	**Robotic Operating Systems (ROS)** ROS ecosystem, ROS - 2 Architecture, Package Management & Workspace Organization, ROS - 2 Communication—Publisher, Subscriber, Topic, Services, Actions, URDF and Robot Modeling, Simulation with Gazebo, RViz, Foxglove, Sensors integration with ROS2, ROS_control for drive-by-wire systems, ROS2 driver interfaces for motor controllers, Real-time topics: /odom, /tf, /sensor_data—/lidar/points, /cmd_vel, / gps/fix ,/ camera/info.
Textbooks:	
1.	Siegwart, R., Nourbakhsh, I.R., & Scaramuzza, D. Introduction to Autonomous Mobile Robots, MIT Press; 2nd Edition, 2011, ISBN: 978-0262015356.
2.	Paden, B., Čáp, M., Yong, S.Z., Yershov, D., & Frazzoli, E. A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles, Foundations and Trends® in Robotics, Now Publishers; 2016, ISBN: 978-1680831146.
Reference Books:	
1.	Maurer, M., Gerdes, J.C., Lenz, B., & Winner, H. Autonomous Driving: Technical, Legal and Social Aspects, Springer; 1st Edition, 2016, ISBN: 978-366248845.
2.	Sutton, R.S., & Barto, A.G. Reinforcement Learning: An Introduction, MIT Press; 2nd Edition, 2018, ISBN: 978-0262039246.
3.	Rajamani, R. Vehicle Dynamics and Control, Springer; 2nd Edition, 2012, ISBN: 978-1461414322.
4.	Ramkumar, L., & Kumar, A. ROS2 in 5 Days: Programming Robots with ROS2, ConstructSim; 2022.
5.	Fairchild, B. & Harman, J. Programming Robots with ROS: A Practical Introduction to the Robot Operating System, O'Reilly Media; 1st Edition, 2016., ISBN: 978-1449323899
e-Resources	
1.	https://nptel.ac.in/courses/112106298
2.	https://www.udemy.com/course/applied-control-systems-for-engineers-2-uav-drone-control/?couponCode=KEEPLEARNING

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4106	PE	3	--	--	3	30	70	3 Hrs.

POWER ELECTRONIC DRIVES

(For EEE)

Course Objectives: Students will learn about

1.	About selection of Electric Drive and different Starting & Braking methods.
2.	About the dynamics of Drive with mechanical load characteristics and Four-Quadrant operation.
3.	About various Speed control methods for rectifier fed DC Motors.
4.	About chopper fed DC Drives.
5.	About various types of Induction motor Drives with slip power recovery schemes.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Select** suitable converter for motor drives & explore different electric starting and braking methods.	K4
2.	**Analyze** the dynamics of electric drive and **demonstrate** the operation of drive to meet the load requirements.	K4
3.	**Apply** and **analyze** rectifier fed DC drives with continuous current mode operation.	K4
4.	**Apply** and **analyze** chopper fed DC drives with closed loop control.	K4
5.	**Investigate** the control strategies for Induction motor Drives and slip power recovery schemes.	K4

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION TO ELECTRICAL DRIVES** Electrical Drive & its advantages, Parts of Electrical Drives – (Electrical Motors, Power Modulators, Sources and Control Unit), Choice of Electrical Drives, Status of DC and AC Drives, Electric Starting and Braking methods.
UNIT-II (10 Hrs)	**DYNAMICS OF ELECTRICAL DRIVES** Fundamental Torque Equation, Multi-Quadrant Operation of Drive, Equivalent values of Drive Parameters – Loads with Rotational motion & Translational motion, Components of Load Torques, Nature and Classification of Load Torques, Steady State Stability, Load Equalization.
UNIT-III (10 Hrs)	**CONTROLLED RECTIFIER FED DC DRIVES** DC Separately excited & Series motors – Speed - Torque relations & characteristics – Armature voltage, Flux and Armature Resistance control, Single-Phase Fully controlled converter fed DC Motors - Continuous current mode operation - Speed-Torque Characteristics – simple problems.
UNIT-IV	**CONTROLLED CHOPPER FED DC DRIVES**

(10 Hrs)	Choppers – four quadrant operation, Chopper controlled DC separately excited motor and DC series motor – Motoring, Regenerative Braking, Dynamic Braking – Speed-Torque Characteristics, simple problems, Closed-Loop Control of DC Drive.
UNIT-V (10 Hrs)	**CONTROL OF INDUCTION MOTORS** Three - phase Induction Motor Speed-Torque relation & its characteristics, Speed control by AC Voltage Controller, Variable frequency control of an Induction motor (V/f control), Closed-Loop Speed control of Induction motor drive, Slip Power Recovery Schemes – Static Scherbius Drive & Static Kramer Drive.
Textbooks:	
1.	G. K. Dubey, "Fundamentals of electric Drives", Narosa Publishing House, 2nd edition, 2011.
2.	Bimal K. Bose, "Modern Power Electronics and AC Drives", Pearson Education Pvt. Ltd., New Delhi, 2014.
Reference Books:	
1.	Power Electronics: Circuits, Devices and Applications by M.H.Rashid, Third Edition, 2009.
2.	Vedam Subramanyam, "Electric Drives - Concepts and Applications", Tata McGraw-Hill publishing company Ltd., New Delhi, 2009.
e-Resources	
1.	https://nptel.ac.in/courses/108104140
2.	https://nptel.ac.in/courses/108108077

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4107	PE	3	--	--	3	30	70	3 Hrs.

POWER SYSTEM PROTECTION

(For EEE)

Course Objectives: Students will learn about

1.	The need for protection and basic principles of arc interruption.
2.	The principle of operation, constructional features and testing of circuit breakers.
3.	The principle of operation and different types of protective relays.
4.	The applications of electromagnetic relays to power system devices & various devices used for overvoltage protection.
5.	The basics of static and numerical relaying and comparators.

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Illustrate the need for protection, rating of circuit breakers and analyze different voltages due to arc interruption.	K4
2.	Apply the arc quenching methods and testing on various types of circuit breakers.	K3
3.	Illustrate the behavior of different types of electromagnetic relays and compute the operating times by using time-current characteristics	K3
4.	Apply electromagnetic relays to alternator, transformer, feeder and busbar protection and illustrate the operation of various protection devices against over voltages.	K3
5.	Illustrate the principles of comparators, static and numerical relaying.	K3

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION TO PROTECTION & CIRCUIT BREAKERS:** Need for protective systems, Nature and causes of faults, Types and effects of faults, Essential qualities of protection, Switchgear Equipment-Isolating Switches. Formation of arc, Methods of arc extinction, Restriking voltage, Recovery voltage, Rate of Rise of Restriking Voltage (RRRV), Single frequency transients, Resistance switching, Current chopping, Ratings of circuit breakers.
UNIT-II (10 Hrs)	**TYPES OF CIRCUIT BREAKERS AND TESTING:** Principle of operation of circuit breakers, Classification of circuit breakers, Construction and working of Air Circuit Breakers, Air Blast Circuit Breakers, Minimum Oil Circuit Breaker (MOCB), Puffer type SF-6 Circuit Breakers and Vacuum Circuit Breakers, Auto reclosure, Testing procedure and Indirect testing of Circuit Breakers.
UNIT-III (10 Hrs)	**PROTECTIVE RELAYS:** Types of relays, Classification of protective Schemes, Primary and Back-up protection,

	Basic relay terminology. Time–Current characteristics, Calculation of relay operating time- Simple problems, Principle of operation & construction of induction type over current relay, Directional over-current relay, Differential relays (Current balanced protection) & percentage differential relays. Universal torque equation, Distance relays – Impedance, Reactance and Mho relays [only basics].
UNIT-IV (10 Hrs)	**APPARATUS PROTECTION:** Protection of Alternators, Transformers, Single and parallel feeders, Bus-Bar, Lightning Arresters, Metal oxide Surge Arrester, Surge Absorber.
UNIT-V (10 Hrs)	**STATIC AND NUMERICAL RELAYING:** Introduction to Static relaying – Advantages and disadvantages of static relays, Basic Block Diagram, Comparators: Phase comparator – Phase splitting type, Integrating type, Amplitude comparator – Rectifier Bridge type, Phase splitting type Numerical Protection: Block diagram of a typical numerical relay, Advantages and disadvantages of numerical relays, Block diagrams of Numerical over current, distance, differential relays.
Textbooks:	
1.	Power System Protection and Switchgear by Badri Ram and D. N. Vishwakarma, Tata McGrawHill Education, Second Edition, 2017.
2.	Principles of Power Systems by V. K Mehta and Rohit Mehta, S. Chand publications, Third edition, 2005.
Reference Books:	
1.	Protection and Switchgear by Bhavesh R. Bhalja R. P. Maheshwari Nilesh Chothani , Oxford University Press, Second Edition, 2018.
2.	A Course in Power Systems by J.B Gupta, S.K Kataria and Sons, 2013.
e-Resources	
1.	https://nptel.ac.in/courses/108107167
2.	https://nptel.ac.in/courses/108105167

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4108	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ADVANCED CONTROL SYSTEMS

(For EEE)

Course Objectives: Students will learn about

1.	The design of basic compensators using Root-locus Plot.
2.	The design of basic compensators using Frequency response.
3.	The Principles of State Feedback controllers.
4.	The use of Digital Computer for Control Systems.
5.	The Stability and Response of the Digital Control Systems.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** Root locus technique for designing basic compensators.	K4
2.	**Apply** Bode plot technique for designing basic compensators.	K4
3.	**Design** State feedback Controllers.	K4
4.	**Illustrate** Modeling of Digital Control Systems.	K3
5.	**Analyze** stability and Response of Digital Control systems.	K4

SYLLABUS

UNIT-I (10Hrs)	**CONTROL SYSTEM DESIGN USING ROOT LOCUS:** Introduction, Improving Steady-State Error Via Cascade Compensation - Ideal Integral Compensation (PI)- Lag Compensation, Improving Transient Response Via Cascade Compensation - Ideal Derivative Compensation (PD), Lead Compensation, Improving Steady-State Error and Transient Response- PID controller Design, Lag-lead-compensator Design, Introduction to Realization of Compensating Networks
UNIT-II (10 Hrs)	**CONTROL SYSTEM DESIGN USING FREQUENCE RESPONSE:** Introduction, Transient Response via Gain Adjustment using Bode Plots, Lag Compensation-Design Procedure, Lead Compensation- Design Procedure, Lag-Lead Compensation - Design Procedure.
UNIT-III (10 Hrs)	**CONTROL SYSTEM DESIGN IN STATE SPACE**: Introduction, State Feedback Controller Design, Pole Placement for plants in Phase-Variable Form, Need for Controllability and Observability, Steady-State Error Design via Integral Control.
UNIT-IV (10 Hrs)	**INTRODUCTION TO DIGITAL CONTROL SYSTEMS**: Introduction, Advantages of Digital Computers, Analog-to-Digital Conversion, Digital to Analog Conversion, Modeling the Digital Computer, Modeling the Sampler, Modeling the Zero Order-Hold, Pulse Transfer Functions, Block Diagram Reduction.

UNIT-V (10 Hrs)	**STABILITY AND RESPONSE OF DIGITAL CONTROL SYSTEMS**: Digital System stability via the Z-Plane, Bilinear Transformations, Jury's Stability test, Steady State Errors for Digital Systems, Transient Response on the Z-plane.

Textbooks:	
1.	Norman S. Nise, 'Control Systems Engineering', Wiley publications (8th Edition).
2.	I. J. Nagrath and M. Gopal, 'Control Systems Engineering', New Age International Publishers (6th Edition).
3.	K. Ogata, Discrete-Time Control Systems, 2nd ed. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.
4.	B. C. Kuo, Digital Control Systems, 2nd ed. New York, NY, USA: Oxford University Press, 1992.
Reference Books:	
1.	Richard C. Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers (8th Edition)
2.	C. H. Houpis and G. B. Lamont, Digital Control Systems: Theory, Hardware, Software, 2nd ed. New York, NY, USA: McGraw-Hill, 1992.
e-Resources	
1.	http://www.digimat.in/nptel/courses/video/108103007/L24.html
2.	https://nptel.ac.in/courses/108103008

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4110	**SEC**	**--**	**1**	**2**	**2**	**30**	**70**	**3 Hrs.**

ELECTRIC VEHICLES LABORATORY

(For EEE)

Course Objectives: Students will learn

1	The basic components of EV.
2	Different types of converters and PWM techniques used for EV.
3	Modes of Charging in EVs
4	Battery performance in EVs
5	Speed control of PMSM used in EV applications.

Course Outcomes: Students will be able to experimentally

S.No	Outcome	Knowledge Level
1	Analyze the performance of electric vehicles (EVs).	K4
2	Explore the working of different types of converters used for EVs.	K4
3	Analyze the performance of EV charging system.	K4
4	Evaluate the battery performance of EVs	K4
5	Explore the speed control of PMSM motor.	K4

LIST OF EXPERIMENTS

Simulation:

1	Simulation of Vehicle Dynamics for Electric Vehicle Applications
2	Simulation of Passive Cell Balancing for a Lithium-Ion Battery Pack
3	Simulation of Dual Active Bridge.
4	Simulation of EV charging using single phase AC supply
5	Simulation of Closed loop speed control of induction motor / BLDC motor.

Hardware:

6	E-Vehicle two-wheeler complete set up.
7	Bidirectional power converter for electric vehicles using ARM Cortex-M4 Microcontroller.
8	Experimental Verification of Controlled DC Output Using a Three-Phase PWM Rectifier.
9	Design of 3S3P Battery Pack with BMS Protection.
10	Estimation of SOC/SOH of EV Battery Systems.
11	Performance analysis of PMSM/BLDC Motor drive.

Add on:

1	Simulation of Lead-acid and Lithium-ion battery.

Reference Books:

1	James Larmine, John Lowry, “Electric Vehicle Technology Explained” Wiley publication, 2ndEdition,2012.

2	M. Ehsani, Y. Gao, S. Gay and A. Emadi, “Modern Electric, Hybrid Electric, and Fuel Cell Vehicles” CRC Press, 2010.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4102	MC	2	--	--	--	30	--	3 Hrs.

TECHNICAL PAPER WRITING & IPR

(Common to AI&DS, CSE, AIML, CSIT, IT, CSD, CSBS, CIC, CE, ME)

Course Objectives:

1.	To appreciate the difference in English used in Academic, Business, Legal and other contexts.
2.	To know the fundamentals of basic technical report structure and writing.
3.	To understand the filing and processing of patent application.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Construct** grammatically sound and concise technical write-ups.	K3
2.	**Prepar**e the outline and structure of a technical paper with essential sections.	K3
3.	**Develop** a project proposal and dissertation framework aligned with academic conventions.	K3
4.	**Use** a word processor effectively for document formatting, citations, and version control.	K3
5.	**Identify** appropriate IPR mechanisms for protecting various types of intellectual creations.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** An introduction to writing technical reports, technical sentences formation, using transitions to join sentences, Using tenses for technical writing. Planning and Structuring: Planning the report, identifying reader(s), Voice, Formatting and structuring the report, Sections of a technical report, Minutes of meeting writing.
UNIT-II (10 Hrs)	**Drafting report and design issues**: The use of drafts, Illustrations and graphics. Final edits: Grammar, spelling, readability and writing in plain English: Writing in plain English, Jargon and final layout issues, Spelling, punctuation and Grammar, Padding, Paragraphs, Ambiguity.
UNIT-III (10 Hrs)	**Proofreading and summaries:** Proofreading, summaries, Activities on summaries. Presenting final reports: Printed presentation, Verbal presentation skills, Introduction to proposals and practice.
UNIT-IV (10 Hrs)	**Using word processor:** Adding a Table of Contents, Updating the Table of Contents, Deleting the Table of Contents, Adding an Index, Creating an Outline, Adding Comments, Tracking Changes, Viewing Changes, Additions, and Comments, Accepting and Rejecting Changes, Working with Footnotes and Endnotes, Inserting citations and Bibliography, Comparing Documents, Combining Documents, Mark documents final and make them read only., Password protect Microsoft Word documents., Using Macros

UNIT-V (10 Hrs)	**Nature of Intellectual Property:** Patents, Designs, Trade and Copyright. Process of **Patenting and Development:** technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property

Textbooks:	
1.	Kompal Bansal &Parshit Bansal, "Fundamentals of IPR for Beginner's", 1st Ed., BS Publications, 2016.
2.	William S. Pfeiffer and Kaye A. Adkins, "Technical Communication: A Practical Approach", Pearson.
Reference Books:	
1.	Ramappa,T., "Intellectual Property Rights Under WTO", 2nd Ed., S Chand, 2015.
2.	Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011.
3.	Day R, How to Write and Publish a Scientific Paper, Cambridge University Press(2006)
e-Resources	
1.	https://www.udemy.com/course/reportwriting/
2.	https://www.udemy.com/course/professional-business-english-and-technical-report-writing/
3.	https://www.udemy.com/course/betterbusinesswriting/

Regulation: R23	**IV / IV - B.Tech. II - Semester**								
ELECTRICAL & ELECTRONICS ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23EE4201	Project and Internship	**PR**	-	-	24	12	60	140	200
		TOTAL	**0**	**0**	**24**	**12**	**60**	**140**	**200**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EE4201	**PR**	**--**	**--**	**24**	**12**	**60**	**140**	**3 Hrs.**

INTERNSHIP & PROJECT WORK

(For EEE)

Course Objectives: Students will learn about

1.	To survey the literature on the selected problem and identify the objective.
2.	To identify the necessary tools and components to initiate the project.
3.	Design, build and test a system with hardware /software.
4	To develop a prototype/model of the project work.
5	To prepare documentation in standard format and communicate technical concepts

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Identify the objectives from the gaps through literature survey and propose solution for solving objective	**K4**
2.	Identify the required tools & components to initiate the project/process at the laboratory level.	**K4**
3.	Design solutions to complex electrical engineering problems utilizing software/hardware approach.	**K5**
4.	Develop the project within the available resources, in stipulated time and with ethical values & social responsibility	**K5**
5.	Write the documentation in standard format and communicate proposed solution orally in a professional manner enhancing self-study and lifelong learning abilities.	**K6**

SYLLABUS

The object of Project Work is to enable the student to take up investigative study in the broad field of Electrical and Electronics Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a supervisor. This is expected to provide a good initiation for the student(s) in R&D work.

The assignment to normally include:

a) Survey and study of published literature on the assigned topic.
b) Working out a preliminary approach to the problem relating to the assigned topic.
c) Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility.
d) Preparing a written report on the study conducted for presentation to the department.
e) Final Seminar, as oral Presentation before a departmental committee.